Question 4 - Install and configure Git and Jenkins

Question 4: Install and configure Git on Ubuntu Linux, verify the installation, and set up global Git configuration (user name and email).

Repository: https://github.com/ksamirdev/ensp461-devops-lab1

Environment

The lab host is an Ubuntu 24.04 Docker container booted with systemd, so Jenkins and the application run as real services. The container holds no GitHub credentials, so pushes were sent from the host machine using its existing GitHub login, while Jenkins pulls from the public repository.

Commands and output

```
$ cat /etc/os-release | head -2
PRETTY_NAME="Ubuntu 24.04.5 LTS"
NAME="Ubuntu"

$ sudo apt-get update -qq

$ sudo apt-get install -y -qq git

$ git --version
git version 2.43.0

$ git config --global user.name 'Samir Kumar Gupta'

$ git config --global user.email 'samirdiff@proton.me'

$ git config --global init.defaultBranch main

$ git config --global --list
user.name=Samir Kumar Gupta
user.email=samirdiff@proton.me
init.defaultbranch=main
```

Fig: Git install, version check and global configuration (rendered from evidence/logs/q4-install.log)

```
$ curl -fsSL https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key | sudo tee
/usr/share/keyrings/jenkins-keyring.asc >/dev/null

$ echo 'deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] https://pkg.jenkins.io/debian-stable
binary/' | sudo tee /etc/apt/sources.list.d/jenkins.list
deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] https://pkg.jenkins.io/debian-stable binary/

$ sudo apt-get update -qq

$ sudo apt-get install -y -qq jenkins
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package net-tools.
Preparing to unpack .../net-tools_2.10-0.1ubuntu4.4_arm64.deb ...
Unpacking net-tools (2.10-0.1ubuntu4.4) ...
Selecting previously unselected package jenkins.
Preparing to unpack .../jenkins_2.580.2_all.deb ...
Unpacking jenkins (2.580.2) ...
Setting up net-tools (2.10-0.1ubuntu4.4) ...
Setting up jenkins (2.580.2) ...
invoke-rc.d: policy-rc.d denied execution of start.
Created symlink /etc/systemd/system/multi-user.target.wants/jenkins.service →
/usr/lib/systemd/system/jenkins.service.

/usr/sbin/policy-rc.d returned 101, not running 'start jenkins.service'

$ systemctl is-active jenkins
inactive

$ dpkg -s jenkins | grep -E '^(Package|Version)'
Package: jenkins
Version: 2.580.2

$ sudo ss -ltnp | grep 8080
```

Fig: Jenkins repository key, install and package version (rendered from evidence/logs/q4-install.log)

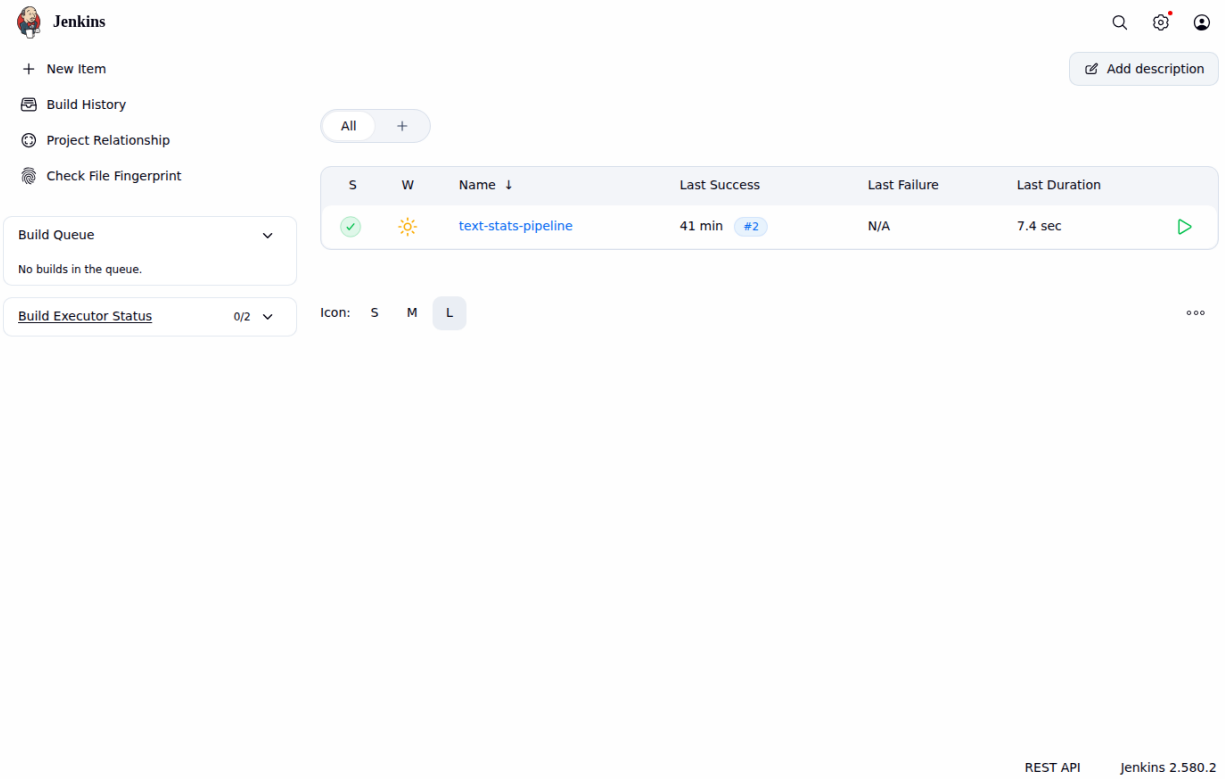


Fig: Jenkins dashboard after setup

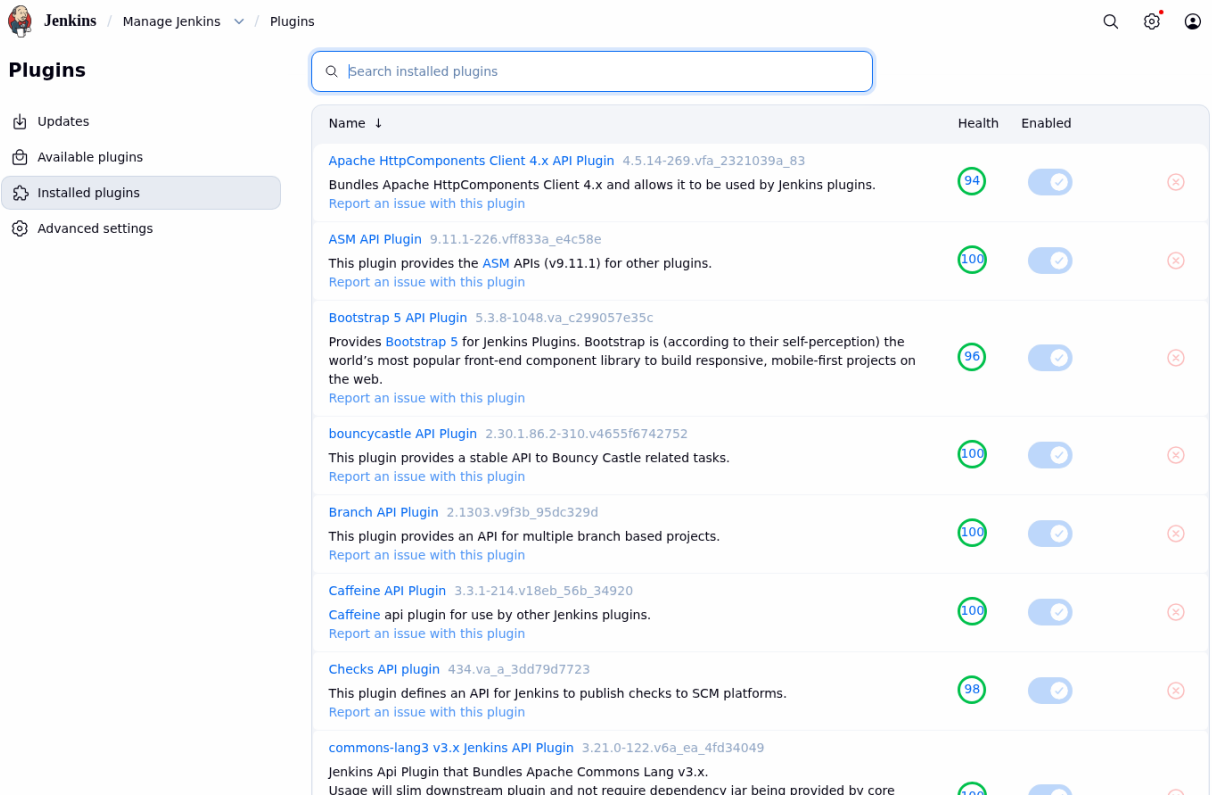


Fig: Installed Jenkins plugins

Observations

- Git and Jenkins both installed from apt; the container's minimal image also needed python3-venv, curl and fontconfig.
- Global identity set with git config lives in ~/.gitconfig and applies to every repository for that user.
- Jenkins 2.580.2 was installed with OpenJDK 21, using the 2026 Jenkins repository signing key.
- The apt post-install step did not start Jenkins inside the container (policy-rc.d denied it), so the service was started with systemctl.